

LLM Architecture: Embeddings & Attentions & Local LLM

Embeddings: textlarni manosi saqlab qolgan holda ularni raqamga utkazish usul. LLM suzlarni matnlarni tushunmaydi, shunga high-dimensional dan low-dimensional vectorlarga utkazvolamiz.

Modelni dimensionaliga qarab convert bulingan raqamlarni soni harhil buladi. misol, $d_model = 4$ - [0.2, 1.4, -0.8, 1.2] – shu vectorlarni masofasiga qarab objectlar qanchalik bir biriga yaqinligini bilsak buladi.

Data type - Floating Point

Distributed Representation: vectordagi bitta qiymat alohida mano bermaydi, barcha qiymatlar birgalikda objectning manosi ifodalaydi

Embeddings ga uxshaydigon usulni oldin urgangan edik – **Encoding** (one-hot, label etc.)

Feature	Encoding	Learned Embeddings
Definition	Manosiz shunchaki raqamga convert	Manoga karab, uzi urganadi.
Creation	Insondan yozilgan formulalar orqali	Modelni katta textlar tuplamida uqitish orqali buladi
Mano darajasi	Mano yuq. Shunchaki raqam	Chuqur mano bor. Suzlar orasidagi semantic bogliqlik
Vector type	Sparse vector. Asosiy 0, 1	Dense vector. Floating point
Dimensionality	Lugat hajmi oshsa vector ham kengayadi.	Fixed size. Odatda 3072 yoki 4096 ulcham buladi

[Embedding turlari]:

1. Input

Type: Token

Status: Standard (Llama 3, Gemini)

Core concept: Tokenning faqat lugaviy manosi. Matndagi tartibni bilmaydi.

Type: RoPE (Rotary)

Status: Standard (Llama 3, Gemini)

Core concept: vectorlarni burchakka aylantirib, textdagi dagi positionni bildiradi

Structured-based

2. Output

Type: contextual

Status: Modern (GPT, Gemini)

Core concept: Attention mexanizmlar orqali suz raqamlarining contextga qarab
Uzgarishi

3. Historical

Type: static

Status: Outdated (Word2Vec, GloVe). Eskirgan.

Core concept: bitta suz – bitta vector usulida olasidi. Yani butunlay gapni tushunmasdi

Nima uchun RoPE ni ishlatamiz?

1. **Sanoat standard**: barcha modern open-source LLM lar RoPE ishlatadi. Amaliy loyihalarda eng kup uchraydigan mavzu.
2. **Long-context**: 128K+ contextni tushunish uchun RoPE asosini bilish kerak
3. **Relative + Absolute hybrid**: RoPE ikkalasining kuchli tomonlarini birlashtiradi
4. **Extrapolation problem**: RoPE ning "base frequency" parametri qanday sozlanishi context-length kengayishiga bevosita tasir qiladi. fine tuning va model serving ishlarida muhim.

Attentions

Attention: modelning hozirgi tokenni tushunish uchun boshqa tokenlarga turli darajada etibor berish mechanism. Yani, contextga qarab muhim tokenlarga etibor berish usul.

Input embeddings da attentiondan foydalanamiz. Agar bu bulmasa Transformer ham bulmaydi. Natijada suzlar bir-biri bilan ulanmagan holda buladi.

✗ Manosiz vectorlarni yani embeddings ni manoli vectorlarga aylantiradi.

Attention bilan Embeddings birgalikda ishlaydi. Position embedding (RoPE) orqali input sifatida vectorlarni (raqamga convert bulingan vectorlar) manosini, positionini Attention ga yuboradi. Keyin attention ularni mix qilib, suzlarni bir-biriga qaratib manosini topadi.

Attention - tez, arzon, kup hajmda uqiy olish hususiyatlariga ega bulishi kerak.

har bir oldingi malumotni qaytadan uqib chiqish kerak a bu ortiqcha token va vaqt isrof. Buni oldini olish uchun ularni eslab qolish kerak.

Q – Query

K – Key

V – Value

Attentionda shular asosiy 3 lik hisoblanadi.

Attentionni 2 ta turlari bor:

1. **Architectural types** – qisqaroq textlar bilan ishlaydi. Uzini ishlatsak buladi

- **MHA** (Multi-head attention): aniq lekin ogir
- **MAQ** (Multi-query attention): balanced
- **GQA** (Grouped-query attention): tez lekin unchalik aniq emas

2. **Context optimization types** – uzun textlar bilan ishlash uchun. Hohlasak qushamiz.

Local LLM

Local LLM: uzimizni computer yoki serverimizda ishlaydigon model

Claude-based LLM: boshqalar ega bulgan computer/server larda ishlaydi

Aspect	Local LLM	Claude-based LLM
Where runs	Uzimizni computerimizda	Cloud serverlarida
Cost	Bepul	Har oy pul tulaydi
Privacy	Maximal level. No leakage	Serverda joylashgan. Havfli bor
Needs internet?	✗	Shart
Setup	Murakkab	Oson
Examples	Llama 3.4, Qwen, DeepSeek etc.	Claude Sonnet 5, Opus 4.8, Fable 5
Quality	Middle. 7B – 400B parameters	Eng yuqori level. Frontier models
Hardware talabi	Yuqori. Kuchli GPU kerak	Kerak emas. Ularni ishi bizniki emas
Customization	Tuliq control	Limited
Updates	Manually	Automatic
Speed	GPU imizga bogliq	Barqaror tez
Best for	Mahfiy loyihalar, offline muhit, experience	Murakkab hollarda, yuqori sifat, tez natija kerak bulsa